

Predicting Peak Performance:

A Statistical Analysis of Major Marathon Winning Times (1897-2018)

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What is a Marathon?

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5K Marathon

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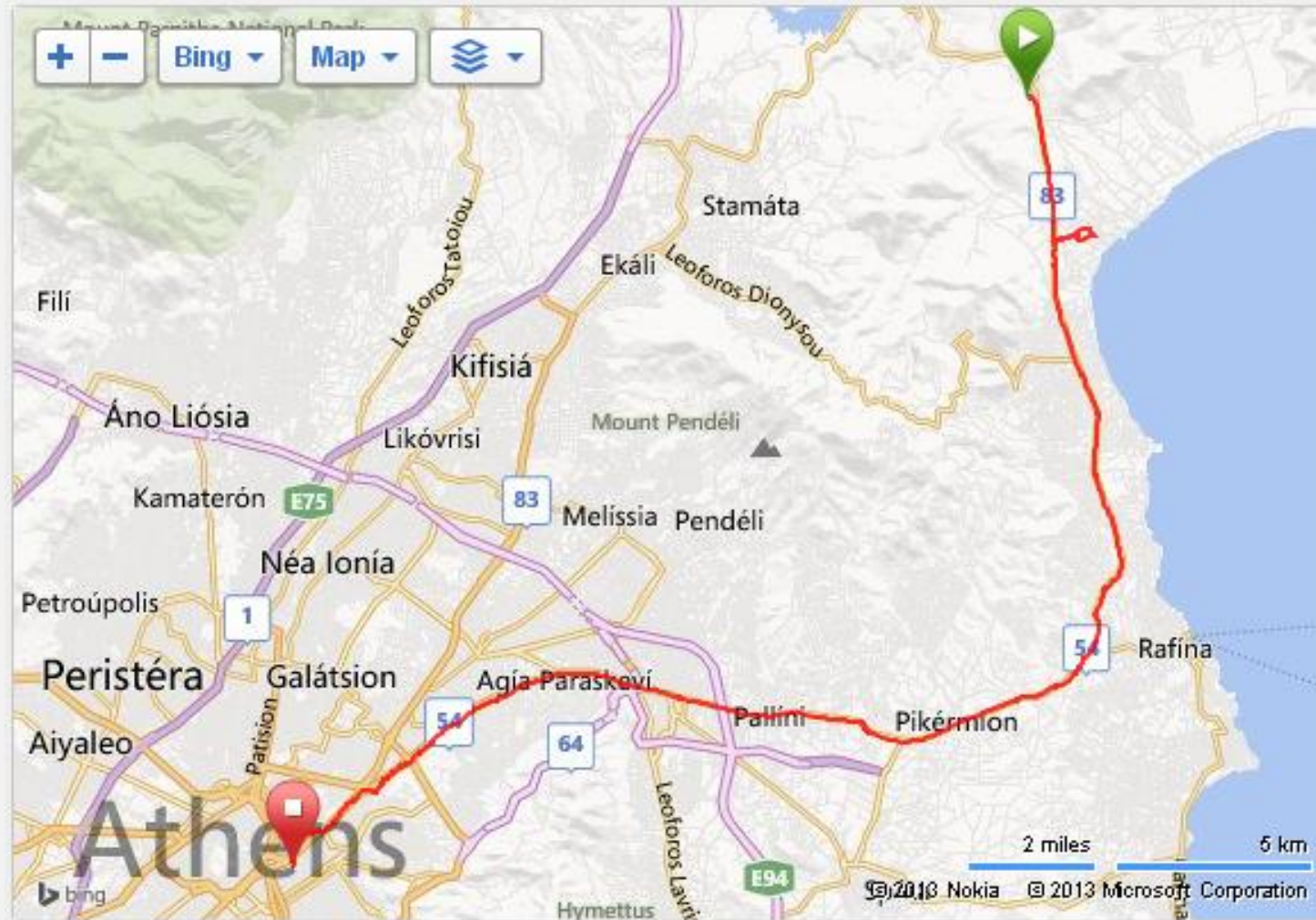
10K Marathon

What is a Marathon?

~~5K Marathon~~

~~10K Marathon~~

What is a Marathon?



42.195km
26.22 miles

The Biological Anomaly



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Initial Data

The **120-Year** Archive: **Male** and **Female** Champions of the **Six** World Marathon Majors

- **Tokyo:** 2007-2018
- **Berlin:** 1974-2018
- **Boston:** 1897-2018
- **New York:** 1970-2018
- **London:** 1981-2018
- **Chicago:** 1977-2018

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- **Missing Variables:** Unrecorded shoe and weather data forced unacceptable levels of mathematical assumption

The New Era Dataset (1980–2018)

Performance	Biometrics	Environmental Factors	Additional information
Time			

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Time	Name		
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Time	Name	Event	
	Gender	Date & Year of Event	
	Age & Year of Birth	Route Feature	
		Weather & Humidity	

The New Era Dataset (1980–2018)

Performance	Biometrics	Environmental Factors	Additional information
Time	Name	Event	Country
	Gender	Date & Year of Event	Shoes
	Age & Year of Birth	Route Feature	
		Weather & Humidity	

Primary Goal

- To build a **predictive** model that identifies the key **biological** and **environmental** factors that define **elite** marathon performance

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Finally, we will translate these **speed predictions back into finish times** to test our accuracy on a blind 20% test set

The OLS Baseline & Its Diagnostic Failure

➤ The Attempt

- **Method:** Forward/Backward Stepwise Regression
- **Initial Variables Kept:** Gender, Country, Route Feature, Shoes, Humidity
- **Baseline Accuracy:** Out-of-Sample RMSE of 0.0634 hours (~3.8 minutes)

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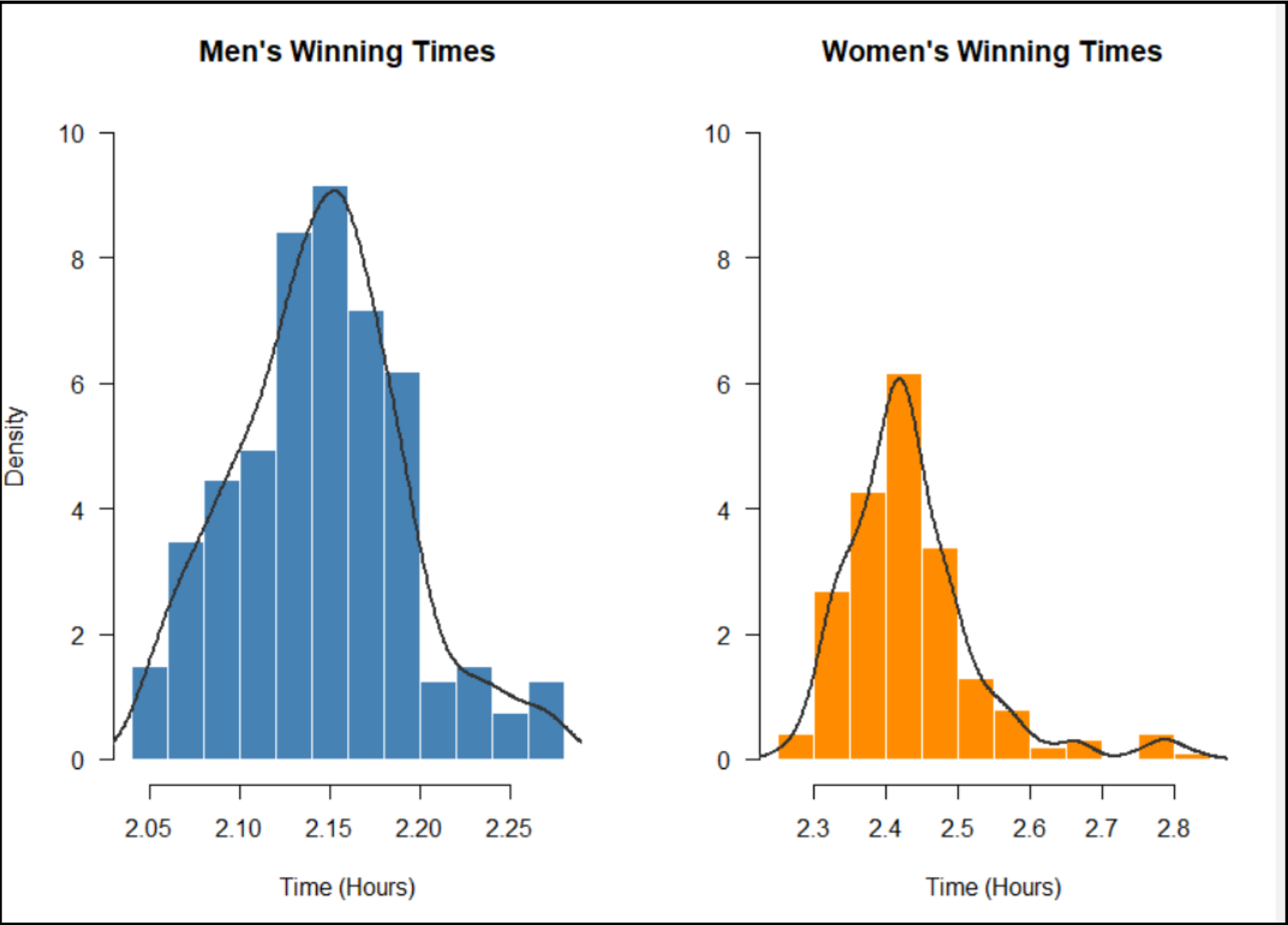
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➤ The Failure

- **Shapiro-Wilk Test** ($p = 2.97e-13$). The physical limit on human speed creates severe **right-skew**
- **Breusch-Pagan Test** ($p = 6.41e-07$). Variance explodes as race conditions worsen (route feature, bad weather)

Biological Wall



GLM Pivot

➤ Solving the Skew

- **The Problem:** OLS failed because elite marathon times are non-negative and right-skewed
- **The Solution:** A Gamma distribution mathematically absorbs the **right-skew** and respects the **biological floor** of the runner

Link Function	AIC	Conclusion
Identity	-893.93	Models raw Time. Sub-optimal fit.
Log	-898.05	Better but not best.
Inverse	-901.65	Models Speed (1/Time). The Winner.

Interactions

- **Forward Stepwise Selection:** We tested **8** potential interaction terms against the base GLM
- **Controlling Complexity:** Variables such as **Name** and **Year of Event** were kept out of the interaction scope to **protect our degrees of freedom** from collapsing
- **AIC Validation:** Terms were only retained if they provided a significant **improvement** in model fit

➤ The Three Key Interactions

- **Gender: Shoes**
- **Gender: Route Feature**
- **Route Feature : Shoes**

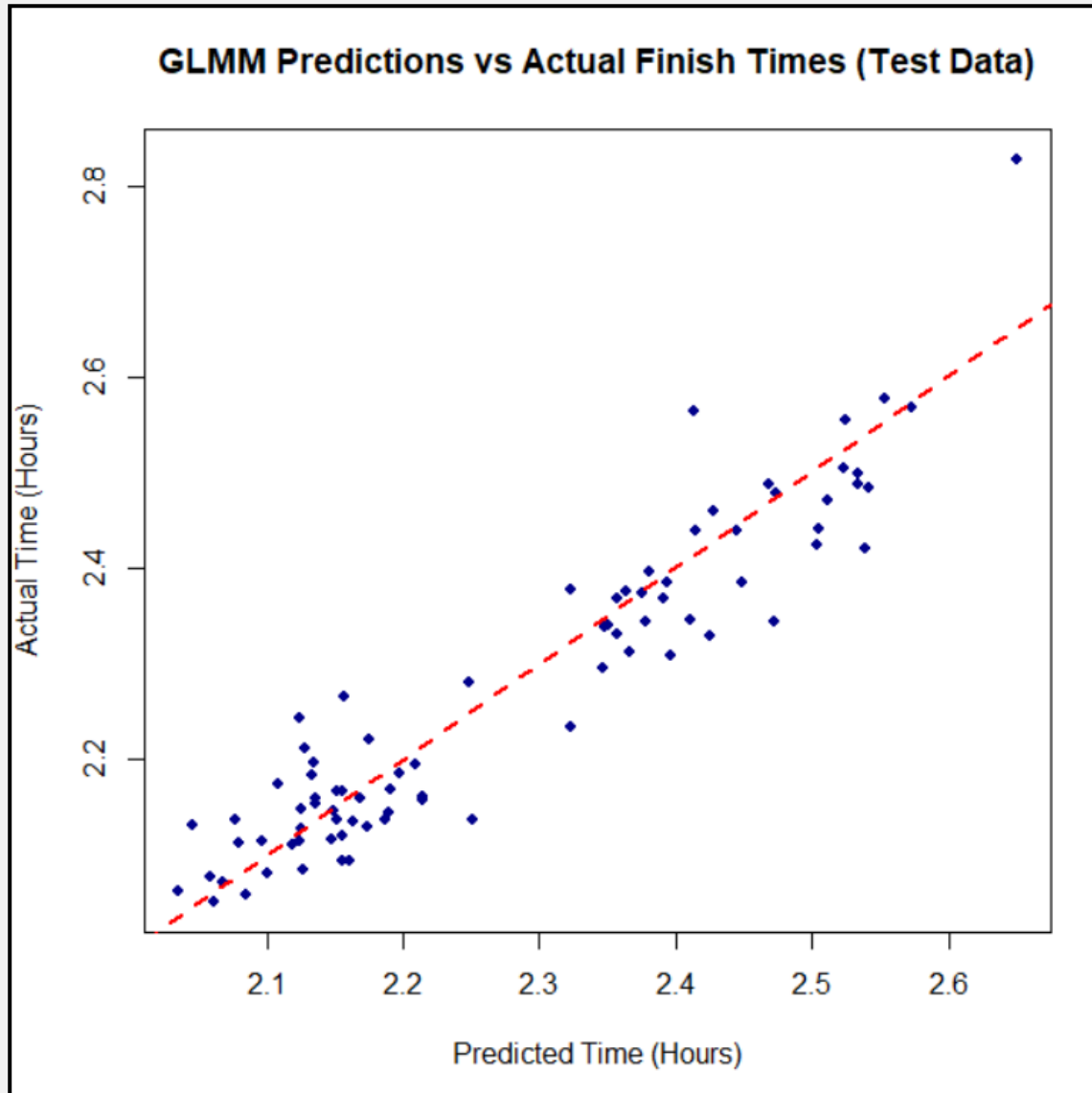
The Pivot: Mixed-Effects model

- **(1 | Name)**: A standard model treats **Eliud Kipchoge** winning five times as five different people. This **random intercept** fixes that by giving him his own personal baseline
- **(1 | Event)**: Captures unmeasured race-day noise, protecting the data from extremes like the **freezing storm of Boston 2007** or the **world-record conditions of Berlin 2014**
- **Scale(Year of Event)**: Prevents chronological confounding. It controls for the natural 40-year **evolution of the sport** (nutrition, training, pacing strategies)
- **ns(scale(Age), df = 3)**: A **3-degree Natural Cubic Spline** to capture the steep climb to an athlete's physical peak and their gradual decline

Key Coefficient Summary

Term	Coefficient (Speed)	p-value	Conclusion
Gender: Male	+ 0.065	< 0.0001	Core Biological Baseline
Shoes: ASICS (vs Adidas)	- 0.015	0.002	Perfect flat route: Adidas > ASICS
Male × Nike	- 0.016	< 0.0001	Higher Energy Return for Females
Male × Numerous Turns	+ 0.006	0.01	Enhanced turning efficiency (musculoskeletal)
Numerous Turns × ASICS	+ 0.016	0.004	Numerous Turns: ASICS > Adidas

The Predictive Proof: Out-of-Sample Accuracy



Key Metrics:

- **RMSE: 0.056 Hours**
(~3.3 Minutes)
- **MAE: 0.0429 Hours**
(~2.5 Minutes)

The Verdict & Next Steps

- **Strategic Footwear Selection:** There is no universally "fastest" shoe. The data proves that **equipment advantage** is dictated by each route. **Adidas** dominates flat straights, while **ASICS** provides advantage on technical courses
- **Gender Factor:** Males maintain a strict **baseline speed** and musculoskeletal advantage, but Carbon-Plate technology (Nike) provides a statistically **higher energy** return for women, likely due to **power-to-weight ratio** optimizations
- **Individualized Bio-Profiling:** Transitioning from macro-level variables to **micro-level biometrics** (VO2 Max, baseline lactate thresholds and advanced blood panels) to mathematically **quantify** the exact **capacity** of each athlete's engine before the starting gun

Thank you for your attention!